

1.1 Digital Computers and Digital Systems

Digital computers have made possible many scientific, industrial, and commercial advances that would have been unattainable otherwise.

Computers are used in scientific calculations, commercial and business data processing, air traffic control, space guidance, the educational fields and many other areas.

The most amazing property of a digital computer is its generality. It can follow a sequence of instructions, called a program that operates on a given data.

The user can specify and change programs and /or data according to the specific need.

As a result of this flexibility, general-purpose digital computers can perform a wide variety of information-processing tasks.

The general-purpose digital computer is best-known example of digital system. Other examples include electronic calculators, digital voltmeters and the like.

Characteristic of a digital system is its manipulation of discrete elements of information.

Such discrete elements may be electric impulses, the decimal digits, the letters of the alphabet, arithmetic operations, punctuation marks, or any other set of meaningful symbols.

The placing next to each other of discrete elements of information represents a quantity of information.

For example, the letters d, o, and g form the word dog. The digits 237 form a number.

Thus, a sequence of discrete elements forms a language, that is, a discipline that conveys information.

Early digital computers were used mostly for numerical computations.

In this case, the discrete elements used are the digits.

From this application, the term digital computer has emerged.

Discrete elements of information are represented in a digital system by physical quantities called signals.

Electrical signals such as voltages and currents are the most common.

The signals in all present-day electronic digital systems have only two discrete values and are said to be binary.

The digital-system designer is restricted to the use of binary signals because of the lower reliability of many-valued electronic circuits.

In other words, a circuit with ten states, using one discrete voltage value for each state, can be designed, but it would possess a very low reliability of operation.

In contrast, a transistor circuit that is either on or off has two possible signal values and can be constructed to be extremely reliable.

Because of this physical restriction of components, and because human logic tends to be binary, digital systems that are constrained to take discrete values are further constrained to take binary values.

To simulate a physical process in a digital computer, the quantities must be quantized.

When the variables of the process are presented by real-time continuous signals, the latter are quantized by an analog-to-digital conversion device.

A physical system whose behavior is described by mathematical equations is simulated in a digital computer by means of numerical methods.

When the problem to be processed is inherently discrete, as in commercial applications, the digital computer manipulates the variables in their natural form.

Block diagram of the digital computer is shown in Fig 1-1.